

# HARIBABU HASINI

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## PROFILE

Software engineer in training with hands-on experience across AI-assisted development, agentic workflows, and data engineering — from building LLM tool-calling systems to end-to-end ETL pipelines. Strong foundation in Python, SQL, OOP, and DBMS, with a focus on writing reliable, production-minded code. Seeking to bring this practical, self-driven experience to a full-time software or AI engineering role.

## EDUCATION

**BTech, Amrita Vishwa Vidyapeetham**  
Electronics and Communication Engineering

Kerala, India | 10/2022 – 07/2026

## SKILLS

**Programming & Analysis:** Python, SQL  
**Core Concepts:** Object-Oriented Programming (OOP), DBMS, Data Structures & Algorithms (DSA)  
**System Architecture:** In-Memory Databases (DuckDB), API Integration (GraphQL/REST)  
**AI & Prompt Engineering:** AI-Assisted Development, Prompt Engineering, Agentic Workflows, LLM Tool Calling (OpenRouter), Read-Only Security Gateways  
**Project & Deployment:** Streamlit Deployment, Version Control (Git/GitHub), QA Testing & Verification, Technical Troubleshooting

## PROJECTS

**Conversational BI Agent for monday.com** ⓘ , *Personal Project*2026

- Built a conversational AI agent (Python, Streamlit) enabling executives to query live monday.com ⓘ boards in natural language and receive verified, board-ready financial figures.
- Engineered a deterministic data-profiling layer that ingests raw GraphQL data into an in-memory DuckDB store, infers semantic column types, and computes conditional null rates — grounding the LLM in real data constraints before any query runs.
- Built a multi-step tool-calling loop (via OpenRouter, with LLM fallback orchestration) that lets the agent propose reporting plans, generate and execute complex SQL, and self-recover from syntax errors.
- Implemented data-quality-aware prompting that forces the LLM to generate dynamic caveats tied to underlying data gaps (e.g., flagging when an aggregate relies on a column that's 60% null).
- Hardened the architecture with a strict, read-only API gateway that structurally blocks unauthorized GraphQL mutations, ensuring safe querying of production data.

**Social Media ETL Pipeline** ⓘ , *Personal Project*2025

Python, SQL, MySQL

- Built an end-to-end ETL pipeline in Python to extract, clean, transform, and load 500 social media posts (via public API) into a MySQL database for structured analytics.
- Modularized the pipeline into separate extract, transform, and load scripts, improving maintainability and making it easier to debug and onboard new data sources.
- Classified posts into 3 sentiment categories, finding the dataset was 91.2% Neutral and 8.8% Negative, with no missing or duplicate values after cleaning.
- Wrote 15+ SQL queries — including sentiment and engagement breakdowns — finding that Negative-sentiment posts had ~5% higher average engagement than Neutral posts, and that medium-length posts outperformed both short and long posts.

## CERTIFICATES

**Deloitte Australia – Data Analytics Job Simulation (Forage):** ⓘ Analyzed a business dataset and delivered a client-ready summary of key findings.

**SQL (Advanced) — HackerRank:** ⓘ Certified proficiency in advanced SQL including complex joins, window functions, subqueries, and query optimization.

## COMMUNITY OUTREACH PROJECTS

**Mental Health Awareness, Karunagapally**2024

Delivered interactive mental health awareness sessions for 10th grade students at NSS HSS Karuvatta, in collaboration with subject experts.